

Arun Prakash S

AI / ML Engineer

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Open to relocation - GCC

PROFESSIONAL SUMMARY

Final-year Integrated M.Sc. student in Computer Science (AI & ML) at Bharata Mata College, Kochi, graduating June 2027. Experienced in building and deploying full-stack AI applications, RAG-based architectures, computer vision systems, and predictive ML pipelines across healthcare, retail, and sports analytics domains. Seeking AI/ML engineering roles to build scalable, production-ready intelligent systems.

EXPERIENCE

Health Care AI Deployment Intern | Genesys AI Labs, Kochi, Kerala | July 2026 - Present

- Assisted in deploying AI-powered patient monitoring systems across hospital environments.
- Supported device setup, testing, onboarding, and deployment of AI healthcare solutions.
- Collaborated with doctors, nurses, biomedical engineers, and hospital administrators to facilitate AI system integration.
- Coordinated with product and engineering teams to troubleshoot deployment issues and ensure smooth implementation.
- Contributed to deployment documentation, validation, and operational support during pilot implementations at leading healthcare institutions.

PROJECTS

PatchTST-Driven Prognostic Health Management Framework for Remaining Useful Life Estimation in Turbofan Engine Systems

Python · PyTorch · PatchTST · SHAP · R · R Shiny · LLM · Predictive Maintenance

- Developed an end-to-end Prognostic Health Management framework using the PatchTST transformer to estimate Remaining Useful Life (RUL) of turbofan engines from the NASA C-MAPSS dataset, achieving RMSE: 9.69, MAE: 6.67, and R^2 : 0.9755.
- Built a complete machine learning pipeline for data preprocessing, feature engineering, model training, evaluation, hyperparameter tuning, and SHAP-based explainable AI.
- Developed an interactive R Shiny dashboard for engine health monitoring, degradation analysis, and live RUL prediction with dynamic visualizations.
- Integrated an LLM-powered maintenance assistant providing prediction explanations, automated maintenance reports, maintenance recommendations, and an AI Engineer Chatbot for decision support.

AtmosMood Pipeline

Python · FastAPI · Docker · Apache Airflow · Pydantic · REST API · Hugging Face Transformers

- Built a containerized AI-powered data pipeline that integrates real-time weather data retrieval and sentiment analysis through RESTful APIs.
- Implemented modular FastAPI architecture with separate routers, services, and Pydantic schemas, enabling request validation and automatic API documentation.
- Integrated the wtr.in API for live weather information and Hugging Face Transformers for AI-based sentiment analysis, with robust exception handling for reliable API responses.
- Designed the backend for scalable integration with PostgreSQL, Apache Airflow, and a React dashboard, following production-oriented software engineering practices.

Daily Labour Finding System (DLFS)

MongoDB · Express.js · React.js · Node.js · REST API · JWT Authentication

- Built a full-stack web platform connecting daily wage workers with employers, enabling seamless job discovery and workforce hiring through a centralized application.

- Implemented secure worker registration, employer authentication, job posting, and job search functionalities using RESTful APIs and JWT-based authentication.
- Developed a responsive React.js frontend with an Express.js and Node.js backend, ensuring efficient data management, scalability, and a smooth user experience across devices.
- Evaluated model performance using RMSE, MAE, and R^2 , demonstrating the effectiveness of transformer-based predictive maintenance.

EDUCATION

Integrated M.Sc. Computer Science (AI & ML) | 2022 – 2027

Bharata Mata College, Kochi, Kerala

Higher Secondary - Computer Science | 2022 | 69%

KKM Higher Secondary School, Vandithavalam, Palakkad, Kerala

CBSE Class X | 2020 | 82%

THS Chittur, Palakkad, Kerala

CREDENTIALS & CERTIFICATIONS

Python Programming | iROHUB Infotech | 2023

- Gained hands-on experience in Python programming, covering core programming concepts and problem-solving.
- Strengthened problem-solving, debugging, and logical thinking skills through practical coding exercises.

TECHNICAL SKILLS

Languages: Python, R, SQL, Java, C, C++, JavaScript

Frameworks & Libraries: PyTorch, Pandas, NumPy, Scikit-learn, R Shiny, React.js, Node.js, Express.js, MongoDB

ML & AI: PatchTST, Time Series Forecasting, Predictive Maintenance, Remaining Useful Life (RUL) Estimation, Supervised Learning, Explainable AI (SHAP), Hyperparameter Tuning, Model Evaluation, Transformer Models, Multivariate Time-Series Analysis, Prognostic Health Management (PHM), LLM Integration.

Web Development & Backend: MERN Stack, REST APIs, JWT Authentication.

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, RStudio, MySQL Workbench, Linux, Docker.

Concepts: Data Visualization, ETL, Feature Engineering, Statistical Analysis, Time Series Analysis, Predictive Analytics, Database Management, Machine Learning Pipelines, Data Preprocessing, Regression Analysis, Model Interpretability, AI-Powered Decision Support

Languages: Malayalam, English, Tamil.

LEADERSHIP & OUTREACH

Media & Discipline Head - International Conference on AI & ML | 2026

2-Day Conference, Kochi, Kerala

- Led the Media and Discipline teams during a 3-day international AI & ML conference, coordinating volunteers to ensure smooth event execution and organized operations.
- Managed event photography, videography, and social media coverage while supervising crowd management, attendee movement, and coordination with organizing committees throughout workshops, keynote sessions, and research paper presentations.

Media Wing Member - College Union | 2025 - 2026

Kochi, Kerala

- Managed media coverage for college events through photography, videography, content creation, and social media updates.
- Coordinated with student committees to promote events, document activities, and ensure timely communication and publicity across digital platforms.

REFERENCES

Dr. John T Abraham

Head of Department, Computer Science — Bharata Mata College

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